

Gautam Bulusu

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PROFESSIONAL SUMMARY

MS Applied ML student at UMD (GPA: 3.7/4.0) with production engineering and ML research experience across fintech, medical imaging, and robotics. Built Python platforms at scale (71 brands, 6,250+ hours saved), clinical CV systems at 92% radiologist accuracy, and quantitative ML models with 35% Sharpe improvement. Ranked top 4.7% globally (solo) in IMC Prosperity 4 algorithmic trading competition against teams of up to 5.

EDUCATION

University of Maryland, College Park Sep 2025 – May 2027

M.S. in Applied Machine Learning · GPA: 3.7 / 4.0

Coursework: Probability & Statistics, Principles of ML, Optimization, Computing Systems, ML Systems, DSA

KJ Somaiya College of Engineering, Mumbai Sep 2020 – May 2024

B.Tech in Information Technology · GPA: 8.54 / 10.0

WORK EXPERIENCE

Reliance Brands Limited | Mumbai Jul 2024 – Mar 2025

Software Engineer Intern

- Architected and shipped an end-to-end Python + Streamlit financial analysis platform as sole developer, replacing fragmented Excel workflows across 71 international luxury brands (Balenciaga, Armani, Zegna) and reducing processing time by 80%.
- Built automated data pipelines eliminating ~6,250 employee-hours of manual work annually across 25 analysts; integrated real-time P&L, IRR, Payback, and Cashflow dashboards consumed daily by C-suite stakeholders including CFOs and CEO.
- Deployed on Azure App Services with REST API integrations, CI/CD pipelines, and enterprise-grade security — designed for reliability across 71 concurrent brand deployments.

Jio Institute – CVMI Lab | Navi Mumbai Jun – Sep 2023

Computer Vision Research Intern

- Sole undergraduate among PhD researchers; designed and trained a clinical-grade scoliosis assessment system using Meta SAM + UNet++ segmentation on 374 3D/4D CT scans, achieving 92% accuracy versus radiologist ground truth.
- Engineered weakly supervised training pipeline cutting annotation requirements by ~60%; automated vertebrae centroid extraction and Cobb angle computation, reducing clinical review time by 30%.
- Owned full ML lifecycle in Linux/PyTorch: DICOM ingestion, preprocessing, model architecture decisions, training, and clinical evaluation against the VerSe 2019-2020 Grand Challenge dataset.

PROJECTS, COMPETITIONS & LEADERSHIP

IMC Prosperity 4 — Algorithmic Trading Competition | Solo (Team: INFERNO) Apr 2026

IMC Trading · 18,800+ teams, 22,600+ participants globally

- Ranked 883rd globally out of 18,800+ teams (top 4.7%) and 239th in the United States as a solo competitor against teams of up to 5; achieved 661st global manual rank.
- Designed multi-factor algorithmic trading system in Python across 5 rounds — implementing EMA-based market making with inventory skew, informed trader signal detection from named bot trade signatures, and Black-Scholes options pricing for riskless chooser arbitrage.
- Discovered intraday regime structure in Round 5 across 50 novel products: identified zero-sum basket design, persistent directional drifts (validated via leave-one-out cross-validation), and position flip patterns capturing both halves of intraday reversals across 30+ products.

VLM Quantization Bottleneck Analysis — Qwen3-VL-8B Jan – May 2026

UMD MSML 605 Computing Systems for ML · A100 GPU Experiments (lead)

- Discovered image resolution — not quantization level — determines the GPU bottleneck:** at 448×448 (196 tokens) the language decoder dominates (mul_mat_q 45% of GPU time); at 768×768 (4,029 tokens) the ViT encoder seizes control (flash_attn_tile 68%), same model and hardware. Confirmed across all 12 Nsight Systems profiling runs.
- Characterized near-quadratic ViT attention scaling ($O(n^{1.9})$):** 20.6× more tokens produced a 333× increase in per-block time (0.261 ms → 86.9 ms), approaching but not reaching the theoretical $O(n^2)$ bound due to tiling optimizations.
- Surfaced A100 PCIe weight-loading bottleneck:** observed only 14–17 GB/s H2D bandwidth against a 2,000 GB/s HBM2e theoretical — under 1% utilized — because llama.cpp streams weights on demand via PCIe. This makes A100

decode throughput nearly quantization-flat (115–124 tok/s, 7.5% Q2–Q8 range) versus M4 Pro's 65% range on unified memory.

Multi-Objective RL for Autonomous Driving — Pareto Set Learning

Jan – May 2026

UMD MSML 604 Introduction to Optimization

- Applied LibMOON EPO solver to highway-env to learn a continuous Pareto front simultaneously optimizing safety (inverse TTC + collision avoidance), speed (deviation from 30 m/s), and comfort (jerk + lateral acceleration) — single preference-conditioned network covers full trade-off space without retraining.
- Designed deep λ -injection MLP: preference vector re-concatenated at every hidden layer with $10\times$ magnitude scaling to prevent gradient washout; lifted L2 separation between corner-preference logits from ~ 0.12 to ~ 1.4 , eliminating policy collapse across all 3 objectives.
- Built distributed training infrastructure: ProcessPoolExecutor (K=16 workers), per-preference Jacobian aggregation, EPO vk crash-override fix that broke a 60% crash-rate spiral and stabilized training within 100 updates.

Market Regime Detection — NLP + Ensemble ML

Sep – Dec 2025

UMD MSML 603

- Designed stacking ensemble (Random Forest + XGBoost) integrating FinBERT-extracted financial news sentiment with 60+ engineered technical indicators (momentum, volatility, macro) to classify S&P 500 market regimes — achieving 67.8% classification accuracy.
- Applied SHAP interpretability analysis revealing sentiment signal contributed 44% of predictive power with 5–10 day advance warning for market stress events; identified top features driving regime transitions across bull, bear, and sideways conditions.
- **Delivered 35% Sharpe ratio improvement and 26% volatility reduction** via dynamic asset allocation driven by predicted regime; backtested across 2,500+ trading days with 29% lower drawdowns over 10 years versus static allocation baseline.

Retinal Disease Classification — Transfer Learning

Jan – May 2023

KJ Somaiya College of Engineering

- Led team of 3 to fine-tune ResNet50V1 and MobileNetV2 on ODIR 5K fundus images for 4-class classification (Normal, Glaucoma, Cataract, Diabetic Retinopathy); achieved F1 of 0.97/0.90/0.88 through domain-guided preprocessing in collaboration with an ophthalmologist.

Team KJSCE Robocon — Vice-Captain & Software Lead

Aug 2021 – Jun 2023

ABU Robocon 2022 · 6th Place Nationally (IIT Delhi) · Team's Highest-Ever Rank

- Led software team of 15+ engineers; fine-tuned YOLOv5 on custom 1,200-image dataset deployed on Raspberry Pi at 15 FPS; implemented Dijkstra path planning, ROS-based autonomous navigation, and Kinect sensor integration on Jetson Nano.
- Mentored 40+ junior engineers on computer vision, data structures, and embedded systems; designed and delivered 2-day robotics workshop for 80 students; judged at World Robot Olympiad West India Regional 2022.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, JavaScript, R, MATLAB

ML / AI: PyTorch, TensorFlow, scikit-learn, XGBoost, FinBERT, SHAP, OpenCV, YOLOv5, SAM, UNet++, Transformers, LLMs, RAG, LSTM, Reinforcement Learning, LibMOON

Cloud & Infra: AWS, Azure, GCP, Docker, REST APIs, CI/CD, Azure App Services, Git, Linux, Microservices

Data: pandas, numpy, SciPy, ETL Pipelines, Time Series, Statistical Modeling, Backtesting, Streamlit, Tableau, Power BI

Robotics: ROS, Jetson Nano, Raspberry Pi, Hailo-8L NPU, Embedded C++, SLAM, Path Planning

CERTIFICATIONS

Deep Learning & Reinforcement Learning Specialization — Coursera (Jul 2024) · Mathematics for Machine Learning Specialization — Coursera (Jan 2023) · Data Structures Specialization — Coursera (Jan 2023)